INCH-POUND
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DETAIL SPECIFICATION

BRAKE, ELECTRO-MECHANICAL, NAVAL SHIPBOARD

This specification is approved for use by Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers high impact (H.I.) shockproof, fail-safe, electrically released, spring-set brakes for shipboard hull and deck machinery applications.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

COMMERCIAL ITEM DESCRIPTIONS

A-A-59125 - Terminal Boards, Molded, Barrier Screw and Stud Types and Associated Accessories

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-S-901 - Shock Tests, H.I. (High Impact) Shipboard Machinery, Equipment, and Systems, Requirements for

MIL-DTL-917 - Electric Power Equipment Basic Requirements

MIL-DTL-2212 - Contactors and Controllers, Electric Motor AC or DC, and Associated Switching Devices

MIL-DTL-15109 - Resistors and Rheostats, Naval Shipboard

MIL-S-22698 - Steel Plate, Shapes and Bars, Weldable Ordinary Strength and Higher Strength: Structural

MIL-S-24093 - Steel Forgings, Carbon and Alloy Heat Treated

Comments, suggestions, or questions on this document should be addressed to: Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard DC 20376-5160 or emailed to CommandStandards@navy.mil, with the subject line "Document Comment". Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

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MIL-S-24235	-	Stuffing Tubes, Metal, and Packing Assemblies for Electric Cables, General Specification for
MIL-C-24707	-	Castings, Ferrous, General Specification for
MIL-C-24707/1	-	Castings, Ferrous, for Machinery and Structural Applications
MIL-C-24707/5	-	Castings, Ductile Iron and Austenitic Ductile Iron

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-108	-	Definitions and Basic Requirements for Enclosures for Electric and Electronic Equipment
MIL-STD-167-1	-	Mechanical Vibrations of Shipboard Equipment (Type I - Environmental and Type II - Internally Excited)
MIL-STD-461	-	Requirements for Control of Electromagnetic Interference Characteristics of Subsystems and Equipment
MIL-STD-810	-	Environmental Engineering Considerations and Laboratory Tests
MIL-STD-889	-	Dissimilar Metals
MIL-STD-1399-300	-	Interface Standard for Shipboard Systems, Section 300, Electric Power, Alternating Current (Metric)
MIL-STD-1399-390	-	Interface Standard for Shipboard Systems, Section 390, Electric Power, Direct Current, (Other Than Ship's Battery) for Submarines (Metric)

DEPARTMENT OF DEFENSE HANDBOOKS

MIL-HDBK-470	-	Designing and Developing Maintainable Products and Systems, Volume I and Volume II
MIL-HDBK-781	-	Reliability Test Methods, Plans, and Environments for Engineering Development, Qualification, and Production

(Copies of these documents are available online at http://quicksearch.dla.mil.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

NAVAL SEA SYSTEMS COMMAND (NAVSEA) PUBLICATIONS

S9074-AQ-GIB-010/248	-	Requirements for Welding and Brazing Procedure and Performance Qualification
S9074-AR-GIB-010/278	-	Requirements for Fabrication Welding and Inspection, and Casting Inspection and Repair for Machinery, Piping, and Pressure Vessels
T9074-BD-GIB-010/300	-	Requirements for Low Alloy Steel Plate, Forgings, Castings, Shapes, Bars, and Heads of HY-80/100/130 and HSLA-80/100

(Copies of these documents are available online at https://nll.ahf.nmci.navy.mil, requested by phone at 215-697-2626, or requested by email at nllhelpdesk@navy.mil. These publications can be located by searching the Navy Publications Index for the TMIN without the suffix.)

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2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

ASTM INTERNATIONAL

ASTM A216/A216M	-	Standard Specification for Steel Castings, Carbon, Suitable for Fusion Welding, for High-Temperature Service
ASTM A757/A757M	-	Standard Specification for Steel Castings, Ferritic and Martensitic, for Pressure-Containing and Other Applications, for Low-Temperature Service
ASTM F1166	-	Standard Practice for Human Engineering Design for Marine Systems, Equipment, and Facilities

(Copies of these documents are available online at www.astm.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.2.

3.2 General requirements. Brakes shall conform to MIL-DTL-917. In the event of conflict, the following order of precedence shall prevail:

a. The requirements of this specification.

b. The applicable requirements of Government specifications referenced herein.

c. Non-Government standards specified herein.

3.2.1 Insulation. Insulation shall be Class B, except that coils may be Class A, B, or H as specified (see 6.2). Direct current (DC) brakes shall be capable of withstanding dielectric tests as specified in MIL-STD-1399-300. Alternating current (AC) brakes shall be capable of withstanding dielectric tests as specified in MIL-STD-1399-300. DC shunt coils not protected by a shunt resistor or other equally effective means shall be capable of withstanding, for a period of not less than 1 second, a dielectric test voltage of 10 times rated circuit voltage plus 1,000 volts. Rated voltage for submarine applications is the nominal voltage.

3.2.2 Painting. All parts not having a corrosion-resisting treatment or not fabricated of corrosion-resisting materials of the types specified in MIL-DTL-917 shall be painted in accordance with MIL-DTL-917. Touching up shall be permitted for marks or scratches due to factory handling.

3.2.3 Bearings, bushings, and pivot pins. Bearings, bushings, and pivot pins shall be of bronze, corrosion-resisting metal, oil-impregnated insert material, or other similar materials. Non-austenitic corrosion-resisting steel may be used for hardened pins. Bushings shall be self-lubricating wherever practical. Rotating anti-friction bearings shall be sealed and shall be selected to result in an L-10 life of not less than 10,000 hours.

3.2.4 Material and components. Material shall be as specified herein. Components that are in or supported by the Federal supply system shall be utilized to the fullest extent possible.

3.2.4.1 Cast iron. Brake wheels may be of cast iron. Ductile iron in accordance with MIL-C-24707/5 may be used for housings, pressure plates, end bells, and other structural parts where suitable.

3.2.4.2 Aluminum. T-6061-T6 and 70xx-T6 series aluminum alloys shall not be used for welded load bearing fittings or components.

3.2.4.3 Mercury. Mercury, in any form, shall not be used.

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3.2.4.4 Magnesium. Magnesium or its alloys shall not be used.

3.2.4.5 Asbestos. Asbestos, in any form, shall not be used.

3.2.4.6 Brittle material. Brittle material is prohibited, except where the Command or Agency accepts its use for a particular application. Brittle material is material showing less than 10 percent elongation in 2 inches for the standard tensile test. For the static loading case (normal ship operation), brittle material is material being used below its nil ductility transition (NDT) temperature as measured by the Naval Research Laboratory (NRL) drop weight test. Charpy V-notch impact strength values may be used as criteria when these have been correlated with drop weight test results.

3.2.4.7 Cast steel. Cast steel to be welded shall conform to MIL-C-24707 and MIL-C-24707/1 (Grade WCA of ASTM A216/A216M or Grade A1Q of ASTM A757/A757M).

3.2.4.8 Forged steel. Forged steel to be welded shall conform to Class H, Type V of MIL-S-24093.

3.2.4.9 Steel plate. High yield strength steel plate shall be in accordance with T9074-BD-GIB-010/300. All other steel plate shall be in accordance with MIL-S-24093.

3.2.4.10 Cadmium plating. Cadmium plating shall not be used, except where a particular application is approved by NAVSEA.

3.2.4.11 Castings. Castings shall be in accordance with S9074-AR-GIB-010/278.

3.2.4.12 Fabrication, welding, and inspection. Fabrication, welding, and inspection shall be in accordance with S9074-AR-GIB-010/278. Where welding and brazing procedures are implemented, they shall comply with the requirements of S9074-AQ-GIB-010/248.

3.2.4.13 Recycled, recovered, environmentally preferable, or biobased materials. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.2.4.14 Dissimilar metals. Coupled dissimilar metals shall be protected against corrosion in a marine environment in accordance with MIL-STD-889.

3.3 Design.

3.3.1 Operation. Brakes shall be either the shoe or disc type as specified (see 6.2), and shall be coil operated. Brakes shall be spring-set and electrically released.

3.3.1.1 Fail-safe operation. The brake shall automatically apply the specified resisting static torque to the connected load when electrical power to the brake is removed.

3.3.2 Torque. The static torque of the brake shall be as specified (see 6.2). The brake hub shall not rotate when the brake is de-energized and the specified static torque is applied to the brake hub.

3.3.3 Speed. Unless otherwise specified (see 6.2), the brake rotating parts shall be compatible with a connected shaft speed of 1,800 revolutions per minute (rpm).

3.3.4 Electric power. AC brakes shall be designed to operate from shipboard AC power having steady state and transient voltage and frequency characteristics as specified in MIL-STD-1399-300. Those DC brakes that operate from shipboard DC power shall have steady state and transient voltage characteristics as specified in MIL-STD-1399-300. Unless otherwise specified (see 6.2), the brakes shall operate at one of the following voltages:

a. AC:
   (1) 110 volts, 60 Hertz, single-phase
   (2) 220 volts, 60 Hertz, single-phase
   (3) 440 volts, 60 Hertz, single-phase

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b. DC:

(1) 115 volts

(2) 230 volts

(3) 250 volts (175 to 355)

3.3.5 Duty. The duty rating for cyclic operation of the brake shall be limited by the thermal capacity (see 6.6.1) of the brake and the maximum allowable coil cycle rate.

3.3.5.1 Thermal capacity. The minimum brake thermal capacity shall be specified (see 6.2) and derived by the following formula for the given application:

$$TC = (E_T * n)/550$$

where (E_T) is the total kinetic energy of all moving elements to be absorbed by the brake and (n) is the required number of stops per minute, not less than 1. Note that (n) cannot exceed the maximum allowable coil cycle rate (see 3.3.5.2). The brake shall be capable of fully absorbing and dissipating this energy with a normal ambient temperature of not more than 22 °C (72 °F). Thermal capacity may be de-rated by 0.007 horsepower seconds per minute (hp-sec/min) per 1 °C (1.8 °F) above normal ambient. Brakes mounted in an orientation 15 degrees or more from horizontal shall be further de-rated.

3.3.5.2 Coil cycle rate. The coil cycle rate limits the number of engagements per minute of a static or holding duty brake. The required coil cycle rate for the application shall be as specified (see 6.2). If not specified, the minimum acceptable rate shall be 12 cycles per minute. The brake coil shall be rated for continuous duty.

3.3.5.3 Electrical operation. The brake shall set and apply the specified static torque within 100 milliseconds of electric power being removed. The brake shall release and allow the hub to rotate freely within 100 milliseconds of electrical power being applied.

3.3.6 Manual release feature. All brakes shall be furnished with a side- or end-mounted, manually operated release mechanism as specified (see 6.2). Access to, and operation of, the brake manual release shall not require the use of tools or removal or disassembly of any part of the brake. The brake manual release shall be hand operated and shall be operable by one person without exceeding the manual force limits of ASTM F1166.

3.3.6.1 Manual release actuator. The brake manual release shall be hand operated through a single handle, knob, or lever. The release actuator shall remain installed on the brake without interference to brake operation or hazard to personnel.

3.3.6.2 Manual release spring return feature. The brake manual release shall be spring returned such that the brake automatically applies the specified static torque to the connected load when manual input to the brake release is discontinued. Unless otherwise specified (see 6.2), the brake manual release shall not be detented or latching.

3.3.7 Enclosures. Brake housing and ancillary enclosures shall be in accordance with MIL-STD-108 and shall be one of the following types, as specified (see 6.2):

a. Drip-proof (45 degrees)

b. Watertight

c. Submersible (15 feet)

d. Explosion-proof

3.3.7.1 Through shaft feature. When specified (see 6.2), the brake shall be designed to allow the load shaft to axially extend completely though the brake assembly.

3.3.7.2 Moisture drain. Watertight and submersible enclosures shall have a moisture drain in the bottom. The opening shall be closed with a ¼-inch or larger pipe plug.

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3.3.8 Mounting method. Brakes shall be motor mounted, gearbox mounted, or bedplate mounted, as specified (see 6.2). The attachment of the brake shall not unduly complicate the removal of the rotor of the motor. Applications requiring the brake to be permanently mounted in an orientation 15 degrees or more from horizontal shall be specified (see 6.2).

3.3.8.1 Shaft interface. The brake hub shall be sized and machined to fit the shaft and key dimensions specified (see 6.2).

3.3.9 Adjustment. The brake shall be self-adjusting to compensate for brake lining wear. Any manual adjustments to the brake shall not require the use of tools not listed in the Federal Supply Catalog (copies of this catalog may be consulted in the office of the Defense Contract Management Agency [DCMA]).

3.3.10 Inspection features. Unless otherwise specified (see 6.2), the brake enclosure (see 3.3.7) shall be provided with indicators, inspection ports, or other features that enable maintenance personnel to determine by inspection if the manufacturer's recommended allowable brake lining wear limit is reached, or the brake is not adjusted to within the manufacturer's recommended tolerances. The inspection features shall allow for these inspections to be accomplished without requiring operation, removal, or disassembly of any load bearing portion of the brake.

3.3.10.1 Brake lining wear indicating switch. If specified (see 6.2), a minimum thickness brake lining switch shall be provided. The switch shall conform to the requirements of MIL-DTL-2212 with normally open and normally closed contacts provided. The contacts shall be rated for the control voltage in accordance with 3.3.4 and specified in 6.2.

3.3.11 Corrosion resistance. The brake shall not be vulnerable to internal corrosion and shall not require internal anti-corrosion preservatives that could interfere with brake performance.

3.3.12 Ambient temperature and humidity. Brakes shall operate as specified herein throughout an ambient temperature range of 0 to 50 °C (32 °F to 122 °F) and an ambient relative humidity range of 5 to 95 percent. For ambient temperatures above 22 °C (72 °F), the thermal capacity (see 3.3.5.1) may be de-rated. Maximum ambient temperature for application shall be as specified (see 6.2).

3.3.13 Temperature rise. Temperature rises shall not exceed the values specified in table I, as adjusted in accordance with 3.3.13.1 through 3.3.13.5.

TABLE I. Maximum temperature rise °C (°F).

Name of part	**Temperature rise (50 °C [122 °F] ambient)**
Shunt coils (AC):	
Class A insulation	60 (140)
Class B insulation	85 (185)
Class H insulation	150 (302)
Shunt coils (DC):	
Class A insulation	55 (131)
Class B insulation	80 (176)
Class H insulation	145 (293)
Bolted connections and terminal connections:	
Not silver plated	50 (122)
Silver plated	65 (149)

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3.3.13.1 Hot-spot temperatures. Table I is based on limiting hot-spot temperatures of 110 °C (230 °F) for Class A, 135 °C (275 °F) for Class B, and 200 °C (392 °F) for Class H insulation.

3.3.13.2 Method of measurement. Shunt coil rises shall be measured by the resistance method. All other rises shall be measured by thermocouple or thermometer.

3.3.13.3 Temperature gradient. If cotton or glass served wire or layer insulation is used, the allowable temperature rise for coils is 5 °C (9 °F) less than that shown in table I.

3.3.13.4 Enclosure differential. Coils in watertight, submersible, or explosion-proof enclosures are permitted an additional 5 °C (9 °F) rise over the values specified in table I.

3.3.13.5 Submarine applications. Coils for 250-volt brakes are permitted temperature rises of 10 °C (18 °F) higher than specified in table I, as modified by 3.3.13.3 and 3.3.13.4.

3.3.13.6 Temperature sensor. If specified (see 6.2), a maximum temperature switch shall be provided and adjusted considering the operability conditions of 3.3.5.1 and table I. The switch shall conform to the requirements of MIL-DTL-2212 with normally open and normally closed contacts provided. The contacts shall be rated for the control voltage in accordance with 3.3.4 and specified in 6.2.

3.3.14 Parts.

3.3.14.1 Coils. Coils for disc brakes shall be encapsulated. Closed terminal lugs with binding screws shall be used for coil wire leads. The minimum wall thickness of encapsulating materials shall be 0.031inch. If the coil is wound on a core tube before encapsulation, the wall thickness may be reduced to 0.016 inch in the core. Coils, where of the removable type, shall be readily removable from the pole or yoke without damage to the coil. The coil shall be rigidly held so as to prevent damage to the coil leads. Coils shall be marked for ready identification.

3.3.14.2 Resistors. Resistors shall conform to MIL-DTL-15109. Resistors shall be so constructed and installed that each unit can be readily replaced. Resistors shall be mounted within the brake or control enclosure. Provision shall be made for readily connecting the leads to the resistor steps and for preventing the leads from coming in contact with the heated resistor.

3.3.14.3 Springs. Springs shall be designed to ensure the proper functioning of the equipment to which they are attached. Springs shall not be stressed beyond their fatigue limits.

3.3.14.4 Terminal leads. All terminal leads shall be made of flexible, stranded, insulated wire compatible with temperatures defined in table I and provided with closed terminal lugs. Polyvinyl chloride (PVC) insulation shall not be used on terminal leads. An insulated sleeve or other suitable means shall be provided to prevent abrasion of the lead insulation by metallic edges of the frame, terminal box, or cable clamp.

3.3.14.5 Terminal boards. The brake shall include a terminal board for connection of the coil leads to all external wiring connections. Terminal boards shall be in accordance with A-A-59125.

3.3.14.6 External cable feed through. Feed through for connections for external cabling shall be in accordance with MIL-S-24235.

3.3.14.7 Wire marking. All wires shall be permanently identified in accordance with MIL-DTL-917 at each termination point by wire markers of synthetic resin tubing or fiber tags. Markers shall identify cable number, wire number, and connection point.

3.3.14.8 Threaded fasteners. All fasteners shall not require the use of tools not listed in the Federal Supply Catalog (copies of this catalog may be consulted in the office of the DCMA). Fastener material selection and interfacing bolted surfaces shall be compatible in accordance with 3.2.4.14. External threaded fasteners shall be provided with a corrosion control method that does not rely on a painted coating.

3.3.15 Operational requirements.

3.3.15.1 Inclination. The brake shall operate without damage or degradation when inclined in any direction at an angle of 45 degrees from the normal mounting position (see 3.3.8).

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3.3.15.2 Vibration. The brake shall operate without damage or degradation when subjected to the vibration requirements of MIL-STD-167-1, Type I and Type II.

3.3.15.3 Shock. Brakes shall meet the requirements of MIL-S-901 for Grade A or Grade B, as specified (see 6.2), in the engaged position for equipment in the stowed configuration. Grade A brakes shall operate satisfactorily before and after the periods of shock. Compliance with this requirement shall not depend upon shock isolation mounts or the use of a device or devices by which the operator can lock the brake in position.

3.3.15.4 Electromagnetic compatibility. The brake shall meet and demonstrate compliance with MIL-STD-461.

3.4 Identification plates. Each brake shall be provided with an identification plate mounted on an exterior part not renewed during its life. Plates shall be brass, corrosion-resistant steel, or anodized aluminum of not less than 0.030 inch in thickness and shall be marked by etching, engraving, stamping, or photo-etch process. Etching, engraving, or stamping shall be not less than 0.003 inch deep. The identification plates for brakes shall be marked to include, as a minimum, the following:

a. Manufacturer's name and CAGE code
b. Manufacturer's model, part number, and serial number
c. Manufacturer drawing number and revision
d. Coil voltage, frequency, current rating and duty
e. Static torque
f. Enclosure type
g. Class of insulation
h. Thermal capacity
i. Assembled weight
j. Rotational inertia

3.5 Workmanship. Neatness, thoroughness, and inspection of welding shall be in accordance with S9074-AR-GIB-010/278. Particular attention shall be given to neatness and thoroughness of marking of parts and subassemblies, support for electric cables, squareness and parallelism of butting joints, and absence of burrs and sharp corners on all parts.

3.6 Human engineering. The brake shall be designed for installation, operation, and maintenance by individuals from the 5th percentile female through 95th percentile male population using the anthropometric and strength requirements of ASTM F1166.

3.7 Reliability. The brake shall not require replacement of component parts, except brake linings, during the mean cycles between failures specified when operated at the duty rate (see 3.3.5).

3.8 Maintainability. Unless otherwise specified (see 6.2), the maximum mean time to repair shall be 4 hours. Verification of proper brake adjustment, inspection for brake wear, and cleaning/removal of dust resulting from brake operation shall not require the use of tools not listed in the Federal Supply Catalog (copies of this catalog may be consulted in the office of the DCMA).

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).
b. Conformance inspection (see 4.3).

4.2 First article inspection. One brake shall be subjected to first article inspection when a first article sample is required (see 6.2). This inspection shall include the examinations and tests as specified in table II.

4.3 Conformance inspection. Each brake shall be subjected to the conformance inspection and testing as specified in table II.

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TABLE II. First article and conformance inspection.

Characteristic	Requirement	Verification	First article	Conformance
Insulation	3.2.1	4.4.1	X	X
Painting	3.2.2	4.4.2	X	X
Bearings	3.2.3	4.4.3	X	X
Material	3.2.4	4.4.4	X	X
Operation	3.3.1	4.4.5	X	X
Fail-safe	3.3.1.1	4.4.5	X	X
Static torque	3.3.2	4.4.6	X	X
Speed	3.3.3	4.4.7	X	
Electric power	3.3.4	4.4.8	X	
Thermal capacity	3.3.5.1	4.4.9	X	
Coil cycle rate	3.3.5.2	4.4.10	X	
Electric operation	3.3.5.3	4.4.11	X	X
Manual operation	3.3.6	4.4.12	X	X
Enclosures	3.3.7	4.4.13	X	
Through shaft	3.3.7.1	4.4.13	X	X
Moisture drain	3.3.7.2	4.4.13	X	X
Mounting	3.3.8	4.4.14	X	
Shaft interface	3.3.8.1	4.4.14	X	X
Adjustment	3.3.9	4.4.15	X	X
Inspection features	3.3.10	4.4.16	X	X
Brake switch	3.3.10.1	4.4.17		
Corrosion resistance	3.3.11	4.4.18	X	X
Ambient temperature/humidity	3.3.12	4.4.19	X	
Temperature rise	3.3.13	4.4.20	X	
Temperature sensor	3.3.13.6	4.4.21	X	X
Coils	3.3.14.1	4.4.22.1	X	X
Resistors	3.3.14.2	4.4.22.2	X	X
Springs	3.3.14.3	4.4.22.3	X	X
Leads	3.3.14.4	4.4.22.4	X	X
Terminal boards	3.3.14.5	4.4.22.5	X	X
Feed through	3.3.14.6	4.4.22.6	X	X
Wire marking	3.3.14.7	4.4.22.7	X	X
Threaded fasteners	3.3.14.8	4.4.22.8	X	X
Inclination	3.3.15.1	4.4.23	X	
Vibration	3.3.15.2	4.4.24	X	
Shock	3.3.15.3	4.4.25	X	
Electromagnetic compatibility	3.3.15.4	4.4.26	X	
Identification plates	3.4	4.4.27	X	X

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TABLE II. First article and conformance inspection – Continued.

Characteristic	Requirement	Verification	First article	Conformance
Workmanship	3.5	4.4.28	X	X
Human engineering	3.6	4.4.29	X	
Reliability test	3.7	4.4.30	X	
Maintainability demonstration	3.8	4.4.31	X	

4.4 Verification procedure.

4.4.1 Insulation. Verify by inspection that appropriate class insulation for application ambient temperature is utilized (see 6.2). Coil insulation resistance shall be measured and corrected to 25 °C (77 °F). Insulation resistance shall be not less than 10 megohms.

4.4.2 Painting. Verify by inspection that all parts not having a corrosion-resisting treatment or not fabricated of corrosion-resisting materials shall have been painted in accordance with MIL-DTL-917.

4.4.3 Bearings. Verify by inspection that all bearings, bushings, and pivot pins are fabricated of corrosion-resistant materials. Verify that bearings are sealed and that, where possible, bushings are also sealed.

4.4.4 Material. Materials of fabrication shall be visually inspected for compliance with requirements of 3.2.4.

4.4.5 Operation. Demonstrate that, when the brake release coil(s) are electrically energized, the brake is fully released. Demonstrate that, when the electrical source is de-energized, the brake is spring-set to engage (fail-safe).

4.4.6 Static torque test. Brakes shall be subjected to a static torque test as follows to verify that the static holding torque delivered by the brake equals or exceeds that specified (see 6.2):

a. Apply a torque equal to the specified static torque (see 6.2) to the brake. Verify the brake hub does not rotate when the specified static torque (see 6.2) is applied.

b. Repeat step 4.4.6.a, but with torque applied in the opposite direction.

c. Manually release the brake, rotate the brake hub 90 degrees, and then reset the brake. Verify the brake hub rotates freely while released.

d. Repeat steps 4.4.6.a and 4.4.6.c three more times to verify static holding torque at each hub rotational position to demonstrate performance and repeatability in each direction of operation.

4.4.7 Speed. Energize brake release coils and demonstrate operation of the brake at maximum speed as specified (see 6.2) for 10 minutes in each direction. The brake shall not display evidence of unusual noise or heat.

4.4.8 Electric power. Those DC brakes that are provided power from shipboard DC power supply shall be tested as specified in MIL-STD-1399-390. Failure of a DC brake to release at 80 percent of rated voltage for DC when cold (room ambient) shall be cause for rejection. AC brakes shall be tested as specified in MIL-STD-1399-300. Failure of an AC brake to release at 85 percent of rated voltage for AC at rated frequency when cold (room ambient) shall be cause for rejection.

4.4.9 Thermal capacity. Verify by calculation that the thermal capacity rating of the brake is suitable for the application considering the maximum kinetic stopping requirement and the maximum number of stops required per minute (see 6.2).

4.4.10 Coil cyclic rate. Verify by inspection that the coil cyclic rating exceeds the application requirement for number of brake cycles per minute (see 6.2).

4.4.11 Electric operation (static torque). The operation of the brake electric release shall be demonstrated as follows:

a. Apply a torque equal to the specified static torque (see 6.2) to the brake. Verify the brake hub does not rotate when the specified static torque (see 6.2) is applied.

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b. Apply electric power of the specified voltage and frequency to the brake coil. Verify that coil energizes and the brake releases. Verify the brake release time does not exceed that specified in 3.3.5.3. Rotate the hub ¼ turn.

c. Remove electric power from the brake coil. Verify the brake resets and the brake hub does not rotate when the specified static torque (see 6.2) is applied. Verify the brake reset time does not exceed that specified in 3.3.5.3.

d. Repeat steps 4.4.11.a through 4.4.11.c nine more times, reversing the direction of rotation each time to demonstrate performance and repeatability in each direction of operation.

4.4.12 Manual release operation. The operation of the brake manual release shall be demonstrated as follows:

a. Apply a torque equal to the specified static torque (see 6.2) to the brake.

b. Operate the manual brake release. Verify that, when operated, the brake manual release allows the input shaft to rotate. Verify that manual input required to operate the brake release does not exceed the force limits specified by ASTM F1166. Rotate the hub ¼ turn.

c. Discontinue manual input to the brake manual release. Verify the brake manual release returns to its normal position, that the brake resets, and that the brake hub does not rotate when the static torque load is applied.

d. Repeat steps 4.4.12.a through 4.4.12.c seven more times, reversing the direction of rotation each time to demonstrate performance and repeatability in each direction of operation.

4.4.13 Enclosures. Enclosures shall be tested and inspected for compliance with the requirements of 3.3.7, as applicable, for the specified degree of tightness (see 6.2). Verify by inspection that, if specified (see 6.2), the through hole provides clearance for the shaft diameter. Verify that the drain plug, if required (see 3.3.7.2), is installed.

4.4.14 Mounting. Verify by inspection that the brake assembly is configured to mount as specified (see 6.2). For brakes that are specified to be mounted at an angle greater than 15 degrees from the horizontal (see 6.2), confirm by inspection that the design will preclude the possibility of dragging friction material when released. Verify by inspection that the brake bore is machined to interface with the shaft diameter as specified (see 6.2).

4.4.15 Adjustment. Verify by inspection that the brake design is provisioned with a self-adjusting mechanism to compensate for brake friction material wear. Also verify that all adjustment points may be performed with the use of tools listed in the Federal Supply Catalog (copies of this catalog may be consulted in the office of the DCMA).

4.4.16 Inspection features. Verify by inspection that the design provides for in-service maintenance inspection of equipment that would require adjustment.

4.4.17 Brake lining wear switch. Verify by demonstration that when the brake lining thickness is below minimum allowable thickness, the switch will actuate.

4.4.18 Corrosion resistance. Verify by inspection that brake interior design and material selection is consistent with corrosion control without the use of materials or coating that could adversely affect brake performance.

4.4.19 Ambient temperature and humidity. Verify by test in accordance with MIL-STD-810, Method 507.4, that the brake design is compatible with the ambient temperature and humidity ranges specified in 3.3.12. Also verify by calculation that if the expected ambient temperature exceeds 22 °C (72 °F), the brake can operate as specified (see 6.2) without exceeding the thermal capability (see 3.3.5.1), or if necessary, the brake shall be de-rated to suit the ambient conditions with NAVSEA approval.

4.4.20 Temperature rise. The temperature rise tests shall be as specified in 4.4.20.1 through 4.4.20.4.

4.4.20.1 General requirement. Temperature rise tests shall be made at rated voltage and frequency with the brake subjected to the duty rating specified in 3.3.5 to determine that the rises specified in table I are not exceeded. Temperature rise tests shall be made on individual parts of the completely assembled equipment. The specified voltage shall be applied to all parts simultaneously so that the mutual heating effect of the several parts will be present.

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4.4.20.2 Starting temperature. Temperature rise tests shall not be started if the ambient differs more than 5 °C (9 °F) from the ambient where the brake has stood during the preceding 2 hours.

4.4.20.3 Ambient variations. A variation of more than 10 °C (18 °F) during a period of 6 hours, or a proportional change for runs of shorter duration, shall in no case be permitted. If the ambient variations are excessive or if the ambient changes rapidly at the end, the test shall be repeated.

4.4.20.4 Duration of tests. The temperature rise test for continuously rated brakes shall be continued until the temperature that can be observed during the test has attained a steady final value. The temperature rise test for intermittently rated brakes shall be continued for the duty cycle specified for the brake.

4.4.21 Temperature sensor. Verify by demonstration that when the maximum allowable operating temperature is exceeded, the switch will actuate.

4.4.22 Parts. Verify by inspection that the following parts are designed and provided.

4.4.22.1 Coils. Verify that the coils on disc brakes are encapsulated. Verify that removable coils can be removed without damage. Verify that the coils are rigidly held in the coil and that the cores are marked for ready replacement identification.

4.4.22.2 Resistors. Verify that the resistors meet the requirements of MIL-DTL-15109, that they are mounted inside the brake housing or control enclosure, and that they are readily replaceable. Verify that the leads are dressed clear of the resistor.

4.4.22.3 Springs. Verify that the springs are not closed coiled or loose.

4.4.22.4 Terminal leads. Verify that the leads are flexible, stranded, insulated wires provided with closed terminal lugs. Verify by inspection that PVC insulation is not used. Protective insulated sleeves shall be provided as necessary to protect the leads from sharp edges.

4.4.22.5 Terminal boards. Verify that a terminal board, in accordance with 3.3.14.5, is provided with coil leads terminated on the board.

4.4.22.6 External cable feed through. Verify that a feed through fitting is provided in accordance with MIL-S-24235.

4.4.22.7 Wire marking. Verify that all internal wiring is permanently identified in accordance with MIL-DTL-917.

4.4.22.8 Threaded fasteners. Verify that all fasteners can be engaged with common shipboard tools as specified in the Federal Supply Catalog (copies of this catalog may be consulted in the office of the DCMA).

4.4.23 Inclination. The brake shall be tested for proper operation in accordance with 4.4.5 at rated voltage and frequency when inclined at 45 degrees in the forward, backward, left, and right positions.

4.4.24 Vibration. The brake shall operate without damage or degradation when subjected to the vibration requirements of MIL-STD-167-1. During the exploratory vibration tests and variable frequency tests, the brake shall be tested in the energized (released) position for not more than 10 continuous minutes or an increase of brake temperature of more than 10 °C (18 °F) above ambient, whichever is longer, while rotating at maximum shaft speed and subjected to the input frequency requirements of Type I testing. The input shaft shall then be stopped and the brake set until the brake has cooled to within 5 °C (9 °F) of ambient. Type I testing shall continue until completion of exploratory vibration tests and variable frequency tests. Endurance testing at selected prominence frequencies shall be conducted with the brake released for 50 percent of the test and engaged for the other 50 percent. Brake temperature shall be limited to an increase of 10 °C (18 °F) above ambient and allowed to cool to within 5 °C (9 °F) of ambient. Type II testing shall be conducted continuously with the brake released. If the brake exhibits a continuous thermal increase during Type II testing, the test shall be stopped and the brake inspected for defects.

4.4.25 Shock. Shock tests shall be conducted in accordance with MIL-S-901 for Grade A or Grade B, as specified (see 6.2).

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4.4.25.1 Rejection criteria.

a. Grade A – Breakage or visible distortion of any parts. Inability to pass the tests specified in 4.4.7, 4.4.11, and 4.4.12 after being subjected to shock.

b. Grade B – Parts of the brake coming adrift, endangering personnel or other equipment.

c. For Grades A and B, any condition that results in an electrical hazard.

d. Failure to pass inspection. The brake shall be disassembled following checks (a) and (b) and inspected thoroughly for damage. The extent of disassembly need be only to the point where the condition of the brake can be easily observed. Any observed effects of the shock shall be recorded.

4.4.25.2 Disposal of shock-tested brakes. Shock-tested brakes shall be disposed of as follows:

a. Brakes which have been subjected to and successfully passed the shock test shall be considered acceptable for shipboard use, provided post-shock tests are passed and corrective measures are taken, and the shock tested article is reconditioned. Deformations which do not cause rejection, but which are in excess of dimensional tolerances specified on the brake drawing, shall be corrected.

b. Grade A brakes which have been subjected to the shock test and have failed structurally will not be acceptable. Grade A brakes which have failed due to fastener, fittings, or other nonstructural parts may be reconditioned and resubmitted for shock tests.

c. Grade B brakes which have been subjected to the shock test and have failed in such a manner that parts have come adrift will not be acceptable.

4.4.26 Electromagnetic compatibility. The brake shall be tested in accordance with MIL-STD-461. When energized, the brake shall comply with the emissions requirements of MIL-STD-461. When subjected to the susceptibility tests of MIL-STD-461, the brake shall not be damaged, rendered inoperable, or exhibit unsafe operation.

4.4.27 Identification plates. Verify by inspection that identification plates are marked as specified in 3.4.

4.4.28 Workmanship. Verify by inspection that any welds are compliant with S9074-AR-GIB-010/278. Also verify that no sharp edges or burrs are present internally or externally, that all fasteners are tight, and all wiring is labeled, dressed, and anchored.

4.4.29 Human engineering. Verify by inspection that the brake design complies with the guidance of ASTM F1166.

4.4.30 Reliability. The reliability of the brake static hold and electric release operation shall be demonstrated by the successful completion of a reliability test using the guidance provided in MIL-HDBK-781. Applicable test level and test plan of MIL-HDBK-781 shall be as specified (see 6.2). The brake shall be cycled without interruption by failure for 250 cycles at a rate of 1 cycle every 3 minutes. Each cycle shall consist of the following:

a. Apply the specified electrical power to the brake. Verify the brake releases and allows the hub to rotate freely at 1800 rpm (±5 percent). Maintain the specified electrical power to the brake for 2 minutes.

b. De-energize the brake. Verify the brake sets and the brake hub does not rotate.

c. After 1 minute, repeat these steps until 250 cycles have been reached.

d. After that 250th cycle, verify the brake sets and the brake hub does not rotate when the specified static torque is applied.

4.4.31 Maintainability. The mean time to repair shall be demonstrated by the successful completion of a maintainability demonstration using the guidance provided in MIL-HDBK-470. As a minimum, the demonstration shall include the following:

a. Inspection and replacement of brake linings.

b. Inspection of brake discs or wheels for wear.

c. Inspection, adjustment, and replacement of coils and springs.

d. Internal cleaning to remove dust generated due to brake lining wear.

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e. Utilization of the inspection features specified in 3.3.10.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The brake is a spring-set, electric coil-released, fail-safe brake intended for shipboard hull and deck machinery applications.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. When first article inspection is required (see 3.1, 4.2, and 6.3).
c. Coil insulation (see 3.2.1 and 4.4.1).
d. Type of brake (see 3.3.1).
e. Static torque (see 3.3.2, 4.4.6, 4.4.11, and 4.4.12).
f. Speed, if other than 1800 rpm (see 3.3.3 and 4.4.7).
g. Supply voltage (see 3.3.4).
h. Thermal capacity (see 3.3.5.1).
i. Coil cycle rate (see 3.3.5.2).
j. Manual release location (see 3.3.6).
k. Detent specification (see 3.3.6.2).
l. Enclosure type (see 3.3.7 and 4.4.13).
m. Through shaft (see 3.3.7.1 and 4.4.13).
n. Type of mounting (see 3.3.8 and 4.4.14).
o. Mounting orientation above horizontal (see 3.3.8 and 4.4.14).
p. Shaft data (see 3.3.8.1 and 4.4.14).
q. Inspection features (see 3.3.10).
r. Whether a brake lining wear switch is to be provided (see 3.3.10.1).
s. Whether brake lining wear switch contacts are to be rated for the control voltage (see 3.3.10.1).
t. Maximum ambient temperature (see 3.3.12 and 4.4.19).
u. Whether a temperature switch is to be provided (see 3.3.13.6).
v. Control voltage to be applied to the temperature switch contacts (see 3.3.13.6).
w. Shock grade (see 3.3.15.3 and 4.4.25).
x. Mean time between failures, if other than 4 hours (see 3.8).
y. Number of stops per minute (see 4.4.9 and 4.4.10).
z. Reliability test (see 4.4.30).
aa. Packaging requirements (see 5.1).

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6.3 First article. When a first article inspection is required (see 6.2), the item should be a first article sample. The first article should consist of one unit. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.4 Sub-contracted material and parts. The packaging requirements of referenced documents listed in Section 2 do not apply when material and parts are acquired by the contractor for incorporation into the equipment and lose their separate identity when the equipment is shipped.

6.5 Subject term (key word) listing.

Coil

Disc

Solenoid

6.6 Definitions.

6.6.1 Thermal capacity. The ability of a brake to absorb and dissipate heat generated solely within the brake without exceeding temperature limitations. Ambient temperature will influence the ability of the brake to dissipate absorbed energy.

6.7 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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ENGINEERING DRAWING TECHNICAL CONTENT

A.1 SCOPE

A.1.1 Scope. This appendix covers information that should be included on the drawings when specified in the contract or order. This appendix is not a mandatory part of this specification. Information contained herein is for guidance only.

A.2 DRAWING CONTENTS

A.2.1 Drawing contents. The drawing should contain the following technical information features:

a. A list of materials, listing all components. Each item should be identified by the part or catalog number assigned by the original manufacturer of that item. Quantity and unit weight of each component should be identified.

b. Assembly and details with front and side view of the brake together with overall dimensions. The center of gravity should be shown in both views of the assembly.

c. Weight of the assembled brake and rotation inertia of brake rotating parts.

d. Type of mounting.

e. Torque rating of the brake.

f. Thermal capacity and coil cyclic rating of the brake.

g. Brake voltage.

h. Ambient temperature rating.

i. Manual release detail.

j. Type of enclosure.

k. Class of insulation.

l. Electric wiring and schematic diagrams.

m. Description of operation.

n. Description of adjustments which can be made on the equipment, including the manufacturer's recommended adjustment values/settings.

o. Static torque verification procedure.

p. Shock qualification letter specifying testing activity and date of approval.

q. Vibration qualification letter specifying testing activity and date of approval.

r. EMI qualification letter specifying testing activity and date of approval.

s. Drawings should be numbered to fit into logical groupings with subassemblies following assemblies, and so forth, starting with the assembly drawing.

t. Drawings should be done on computer aided drafting machines, in a language compatible with the Initial Graphics Exchange Specification (IGES) prepared by the U.S. Bureau of Standards.

u. A table of repair parts and special tools.

v. A table of labels, warnings, cautions, and instruction plates.

w. Specific installation requirements and instructions for the brake being supplied. This should include mounting surface, shaft, and key interface requirements and tolerances, lubrication requirements, and required installation, removal, and maintenance envelopes.

x. Where materials of identical or equal characteristics can be identified by more than one specification or standard, the drawings should reference only one specification or standard in accordance with the following order of preference:

(1) Industry and technical society specification or standard.

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(2) Federal specification.

(3) Military specification.

(4) Manufacturer's specification or standard.

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